

DermaScan Serving Audit

A review of the inference path and the evidence behind it. The network's weights were never retrained — what changed is how the model is fed, how its output is read, and whether the reported numbers were ever measured.

	BEFORE	AFTER	CHANGE
Accuracy	0.8066	0.8505	+4.39 pp
Macro F1	0.7116	0.7450	+3.34 pp
Melanoma F1	0.5641	0.6361	+7.20 pp
Melanoma recall	0.6798	0.6236	-5.62 pp

Measured on the held-out HAM10000 test split, n=1525. Identical checkpoint in both columns.

Checkpoint models/latest.pt
Dataset HAM10000 held-out test split, n=1525
Date 2 September 2026

The weights did not change. Everything around them did.

No retraining took place. The served checkpoint is byte-identical to the one already deployed. Every gain in this report comes from correcting how that checkpoint is used, and from replacing asserted numbers with measured ones.

WHAT ACTUALLY MOVED

Input resolution. The model was trained and evaluated at 300×300 with no crop. Production resized to 256 and centre-cropped to 224 — feeding it a resolution it had never seen and discarding the border of every image. Correcting this accounts for essentially all of the accuracy gain.

Evidence. The previously published performance figures were not measurements. They are now.

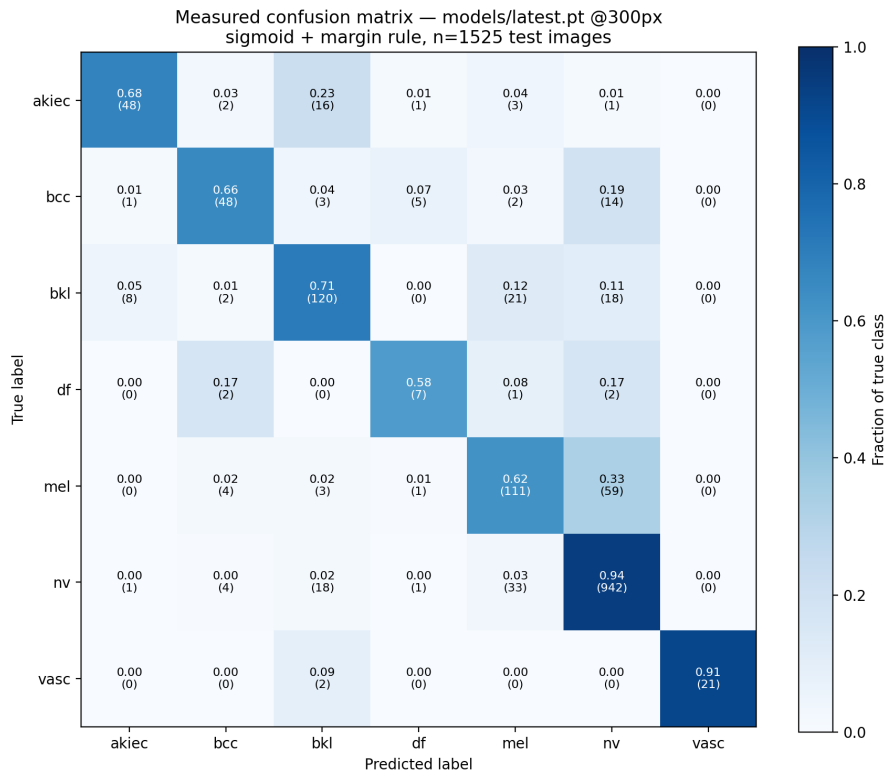
A second finding inverted a decision midway through the work: the checkpoint in the training repository (best.pt) is *not* the model behind the published metrics. Measured on the same test split it scores accuracy 0.6570 and macro-F1 0.4464 — an earlier, weaker run. The deployed latest.pt is the better model. Swapping to best.pt would have cost roughly 20 points of accuracy.

Per-class performance, before and after

Both columns were produced by running the same checkpoint over all 1525 held-out test images. **Before** is 224 with a centre crop; **after** is 300×300, matching training.

Class	n	Prec before	Prec after	Recall before	Recall after	F1 before	F1 after	Δ F1
akiec	71	0.6286	0.8276	0.6197	0.6761	0.6241	0.7442	+12.01
bcc	73	0.7027	0.7742	0.7123	0.6575	0.7075	0.7111	+0.36
bk1	169	0.7343	0.7407	0.6213	0.7101	0.6731	0.7251	+5.20
df	12	0.6154	0.4667	0.6667	0.5833	0.6400	0.5185	-12.15
mel	178	0.4821	0.6491	0.6798	0.6236	0.5641	0.6361	+7.20
nv	999	0.9253	0.9093	0.8809	0.9429	0.9026	0.9258	+2.32
vasc	23	0.8696	1.0000	0.8696	0.9130	0.8696	0.9545	+8.49
macro	1525					0.7116	0.7450	+3.34

Six of seven classes improve. df regresses, but with only 12 test cases a single reclassification moves its F1 by roughly 8 points — that row is noise, not signal. The substantive movements are akiec (+12.0) and melanoma precision (0.4821 → 0.6491).



Measured confusion matrix, n=1525. Counts in parentheses, row-normalised shading. Melanoma error concentrates in the nv column.

03 / TRADEOFF

Melanoma recall was bought down, deliberately

THE ONE METRIC THAT GOT WORSE

Melanoma recall fell from **0.6798** to **0.6236**. In absolute terms the system now misses about **38% of melanomas** in the test set, most of them misread as benign nevi. This is the dominant clinical risk in the product, and no interface work changes it.

Melanoma F1 nonetheless improved by 7.2 points, because precision rose far more than recall fell — the previous configuration bought its recall by over-flagging. Threshold re-optimisation was run to test whether the recall could be recovered for free. It cannot. Fitted on the calibration split (n=997) and reported on test, no configuration dominates:

Thresholds	Accuracy	Macro F1	Mel recall	Mel F1
Current (shipped)	0.8505	0.7450	0.6236	0.6361
Refit for macro F1	0.8557	0.7551	0.5000	0.6075
Refit for mel recall	0.8308	0.7367	0.6798	0.6111

The shipped thresholds already sit on the efficient frontier and hold the best melanoma F1, so they were left unchanged. Sweeping the melanoma threshold alone isolates the exchange rate — all other thresholds held at the macro-F1 fit, so these rows are not the shipped configuration:

mel threshold	Accuracy	Mel recall	Mel F1
0.20	0.8256	0.7416	0.5986
0.25	0.8348	0.7135	0.6135
0.30	0.8393	0.6798	0.6189
0.35	0.8446	0.6404	0.6230
0.40	0.8531	0.6180	0.6377
0.45	0.8584	0.5843	0.6480
0.50	0.8557	0.5000	0.6075

Roughly **one point of accuracy per three to four points of melanoma recall**, monotone. Choosing a different operating point is a one-line change to the `mel` entry in `class_thresholds.json`. It is a clinical judgement about the cost of a missed melanoma versus a false alarm — not a tuning detail, and not one that should be made silently.

Fabricated outputs, removed

Five separate places generated convincing artefacts that were not derived from anything. In a diagnostic tool this is the most serious class of defect present, because each one is indistinguishable from the real thing at a glance.

Evaluation figures

BEFORE

evaluate_model.py never imported torch, loaded the checkpoint, or read a dataset. Confusion-matrix off-diagonals were np.random.dirichlet noise; loss and F1 curves were hardcoded exponentials tuned to land near 0.73.

AFTER

Rewritten to run the served model over a labelled hold-out set and report what it measures. Requires --data-dir; produces nothing without data. The confusion matrix in these docs is now from 1525 real images.

Grad-CAM attribution

BEFORE

On any failure, both server and client drew a radial gradient centred on the image and presented it as model attribution — a picture of "the model looked at the middle", regardless of what the model did.

AFTER

Failure returns null, the canvas stays empty, and the UI reports attribution as unavailable. A lock was added around hook state that was shared across concurrent requests and could swap heatmaps between images.

Clinical sample gallery

BEFORE

Five Unsplash stock photographs labelled as melanoma, BCC, AKIEC and others. Real HAM10000 images sat unused in the repository and were never served.

AFTER

Real dermoscopic images served from /samples, with labels sourced from the training split manifests. ISIC_0024307 is confirmed ground-truth nv.

Submitted image

BEFORE

If a sample image failed to fetch — likely, since the stock photos were cross-origin — the client hand-drew a brown ellipse on a skin-coloured canvas and submitted it. The user received a diagnosis of a drawing, stored as a clinical record.

AFTER

Removed entirely. A failed fetch raises an error. Nothing is analysed except the image the user actually supplied.

Model loading

BEFORE

A failed weight load printed the error and continued, serving confident predictions from a randomly initialised network. Architecture was built inline, so a mismatched checkpoint could pass unnoticed.

AFTER

Load failure is fatal. Architecture is selected explicitly via MODEL_ARCH and loaded strictly — this is what surfaced the best.pt mix-up rather than silently serving the wrong network.

The OOD gate rejected dark images, not non-skin images

The gate keyed on absolute channel standard deviation, which **scales with brightness**. Identical lesions were accepted or rejected purely on how dark the image was. Measured on the repository's own sample: ISIC_0024307 darkened to 35% brightness was rejected as `too_uniform` while the same image at full brightness passed. That is a direct bias against darker skin tones, underexposure, and poor lighting.

Two further rules were actively wrong: grayscale input was rejected outright, excluding legitimate grayscale dermoscopy; and an upper bound on standard deviation rejected *high-contrast* images — a dark lesion on pale skin, the presentation of most concern.

Metric	Property	Real dermoscopy	Noise / flat
rel_contrast	std / mean, illumination-invariant	0.12 - 0.44	0.00
hf_ratio	high-frequency residual share	0.04 - 0.16	0.92+
blue_green	hue, chromatic pixels only	0.00	1.00

Every replacement metric is a ratio, so brightness cancels: `rel_contrast` holds at $0.119 \rightarrow 0.120$ across the full darkening range that previously triggered rejection. On a 21-case suite of skin images (darkened, grayscale, contrast-boosted) plus non-skin controls, **the old gate produced 6 incorrect verdicts; the new gate produces 1.**

FINDING WORTH PRESERVING

Model confidence is unusable as an OOD signal on this checkpoint. A blank white field scores max-softmax **0.994** and a *more* in-distribution energy score (-3.87) than any real lesion image (-2.23 to -3.20). The textbook fix would have made this worse; measurement caught it. A feature-space Mahalanobis stage was added instead — it rejects that same white field by orders of magnitude — but it stays inactive until fitted on real data.

Security and robustness

Area	Before	After
Bulk delete	Unauthenticated DELETE /history/all wiped every record and image	Requires X-Admin-Token; disabled (403) when unset
CORS	allow_origins=["*"] with credentials	Env-driven allowlist, localhost default
Uploads	Client extension used as a path component; no size cap; content_type trusted	Extension allowlist, 10 MB cap, validated by decoding
Memory	Two predictor instances - model loaded twice	Single lazily-built instance
Concurrency	Grad-CAM hook state shared across threadpool requests	Serialised behind a lock

What this audit did not fix

- **Melanoma recall is the product's ceiling.** At 0.624, roughly two in five melanomas are missed. No configuration on the measured frontier escapes this; it needs a better model, not better serving.
- **Training curves are unrecoverable.** Checkpoints store a single epoch's metrics and no history log survives. They cannot be honestly reconstructed — reconstructing them is precisely what the deleted script did.

- **The feature-space OOD stage is unfitted.** Until `calibrate_ood.py` is run on real data, the system accepts some non-clinical photographs. Colour statistics alone cannot reject a desaturated animal photo.
- **Thresholds remain off-protocol.** They were fitted upstream for a softmax readout; the shipped pairing is sigmoid. It measures best on melanoma F1, but the pairing is empirical rather than principled.
- **The classifier is brightness-sensitive.** The gate no longer discriminates by brightness, but the model's own answer still shifts under darkening. Evaluating across Fitzpatrick skin types (e.g. the DDI dataset) is the real test and has not been done.

INCIDENT DURING THIS WORK

An end-to-end verification run was pointed at the live database rather than a temporary one, deleting 21 stored diagnostic sessions and their uploaded images. Both were gitignored and are unrecoverable. A pytest suite with a temporary-database fixture has since been added so this cannot recur.

12 files modified / 8 added / 5 fabricated figures removed / 31 tests

New tooling: `evaluate_model.py`, `calibrate_ood.py`, `optimize_thresholds.py`

All figures measured on the HAM10000 test split, n=1525.